

1. Project Overview

This project analyzes the Zomato Bangalore restaurants dataset for the Alfido Tech Data Science Internship.

The analysis covers the complete data science pipeline:

- * Data loading and initial inspection
- * Data cleaning (null handling, text-to-numeric conversions)
- * Exploratory Data Analysis with 6+ visualizations
- * Machine Learning model (Random Forest Regressor) for rating prediction
- * 5 strategic recommendations for an Alfido Tech-style platform

2. Dataset Summary

Source: kaggle.com/datasets/bhanupratapbiswas/zomato

Records after cleaning: 36957

Columns used: rate, votes, cost, location, rest_type, cuisines, type, city

Key statistics after cleaning:

- * Average Rating : 3.66/5.0
- * Avg Cost (two) : Rs. 564
- * Avg Votes : 232
- * Most common location: BTM
- * Most common cuisine : North Indian

3. ML Model Evaluation

Model : Random Forest Regressor (n_estimators=150)

Features : location, rest_type, cost, votes

Target : rate (restaurant rating 0-5)

Results:

MSE = 0.0182

RMSE = 0.1348

R2 = 0.8918

Top Feature by Importance: votes (57.63%)

4. Key Findings

* Location Hotspots

BTM, Koramangala, and Indiranagar lead in restaurant density. These areas consistently attract high footfall and competitive dining options.

* Rating vs Cost

Premium dining (Rs. 1000+) shows more consistent ratings, whereas budget restaurants have high variance. However, several budget restaurants achieve ratings ≥ 4.0 , indicating perceived value matters more than price alone.

* Votes and Rating Correlation

'Votes' has the highest correlation with 'rate' (corr: 0.40). Popular restaurants sustain better ratings, suggesting a network-effect loop.

* Cuisines Landscape

North Indian, Chinese, and Fast Food dominate cuisine options. Continental and Italian have fewer entries but score higher average ratings, representing high-value niches.

5. Recommendations for Alfido Tech-Style Platform

Recommendation 1: Prioritise High-Density Hotspot Partnerships

Establish restaurant partnerships in BTM, Koramangala, and Indiranagar first. These neighbourhoods deliver maximum reach for minimal acquisition effort.

Recommendation 2: Launch a 'Budget Gems' Content Series

Data shows restaurants with cost < Rs. 400 and rating ≥ 4.0 exist across multiple areas. A curated 'Budget Gems' section drives aspirational discovery and engagement.

Recommendation 3: Gamified Review System

Since vote count strongly predicts rating stability, incentivise users to leave reviews through points, badges, or food vouchers - creating a virtuous engagement loop.

Recommendation 4: Feature 'Online Order' and 'Table Booking' Prominently

Restaurants offering these features attract higher ratings. Highlight these capabilities as trust signals to convert undecided users.

Recommendation 5: Invest in Niche Cuisine Discovery

Continental, Italian, and fine dining cuisines are underrepresented but show higher ratings. A 'Discover Premium' section with curated niche cuisine recommendations can meaningfully differentiate the platform.